

A PROJECT REPORT ON

**AI-Driven System for Crop Health Management
and Yield Estimation Using a Solar-Powered IoT
Sensor Network**

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ABSTRACT

This project presents a solar-powered precision agriculture system that integrates Internet of Things (IoT) technology with machine learning to support data-driven decision-making in farming. The system comprises three functional modules: a crop yield prediction engine tailored for chilli cultivation, a crop recommendation engine, and a plant disease classification system. Environmental and soil parameters are collected through sensors and transmitted to a cloud-based IoT platform for analysis. Machine learning models are used to provide real-time predictions and recommendations based on current field conditions. All modules are deployed on a low-power embedded system powered by solar energy. The proposed system illustrates the potential for integrating AI-driven solutions with IoT infrastructure in real-world agricultural applications. Future enhancements will aim to extend support for multiple crops and improve system resilience in low-connectivity environments.

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LIST OF ABBREVIATIONS

Abbreviation	Illustration
AI	Artificial Intelligence
IoT	Internet of Things
ML	Machine Learning
KNN	K-Nearest Neighbors
RF	Random Forest
ViT	Vision Transformer
CNN	Convolutional Neural Network
MQTT	Message Queuing Telemetry Transport
API	Application Programming Interface
DHT11	Digital Humidity and Temperature Sensor
ESP32	Espressif Systems 32-bit Microcontroller
LCD	Liquid Crystal Display
CSV	Comma Separated Values
SDLC	Software Development Life Cycle
UAV	Unmanned Aerial Vehicle
Wi-Fi	Wireless Fidelity

CHAPTER 1

INTRODUCTION

1.1 Overview

Precision agriculture is a modern farming practice leveraging technology to make data-driven decisions. This project introduces an integrated, solar-powered smart farming solution combining sensor-based data collection, IoT platforms, and AI models to support Indian farmers in improving crop selection, yield estimation, and disease management.

1.2 Motivation

Traditional agriculture often relies on experience and assumptions, leading to inefficiencies. Rising climate variability, soil degradation, and pest outbreaks necessitate the use of smart technologies for sustainable farming. The motivation behind this project is to bridge the gap between farmers and technology through an affordable and intelligent solution.

1.3 Problem Definition and Objectives

Farmers lack real-time insights into crop suitability, expected yield, and plant health, resulting in losses and inefficient resource utilization.

Objectives:

- To develop a crop recommendation system based on soil and weather parameters
- To predict crop yield in real-time using historical sensor data
- To detect diseases in chilli plants using image-based classification
- To enable all functionalities through a solar-powered, standalone embedded system

1.4 Project Scope & Limitations

Scope:

- Chilli plant focus for disease detection

- Field-ready system with sensor input
- Integration with cloud (ThingSpeak IoT)

Limitations:

- Disease and Yield module limited to chilli leaf diseases
- Requires consistent internet for data sync
- Cannot yet handle multi-crop field segmentation

1.5 Methodologies of Problem solving

- Machine learning models: KNN, Random Forest, Vision Transformer
- Sensor-based data acquisition
- Image classification using transfer learning
- Cloud integration with ThingSpeak for IoT monitoring
- Solar-powered embedded deployment

CHAPTER 2

LITERATURE SURVEY

Precision agriculture has become a vital field in modern farming, aiming to optimize agricultural productivity through the application of machine learning, sensor networks, and real-time data analytics. Recent developments emphasize the integration of soil nutrient sensing, environmental monitoring, and artificial intelligence to enhance decision-making for crop selection, yield estimation, and plant health management. This literature review presents an overview of existing approaches aligned with the three core modules of the present system: crop yield prediction, crop recommendation, and disease detection. Additionally, it examines trends in embedded system deployment for real-time agricultural applications.

Crop yield prediction systems have been widely studied using data-driven models trained on environmental and nutrient parameters. Cammarano [1] demonstrated the use of site-specific nitrogen application combined with machine learning algorithms to improve yield forecasting, particularly when soil and weather data were integrated into the model. Similarly, Harish and Devi [2] proposed a smart farming framework that utilized IoT-enabled sensors to collect temperature, soil moisture, and nutrient values, which were then processed through supervised learning models to predict yield with considerable accuracy. Naik et al. [3] further highlighted the suitability of lightweight models such as K-Nearest Neighbors (KNN) for crop yield estimation, particularly in edge-deployed, resource-constrained agricultural settings.

For crop recommendation, machine learning classifiers have proven highly effective in analyzing soil and climatic conditions. Sethi and Lakhina [4] implemented a Random Forest-based crop advisory system using NPK values, temperature, and humidity as inputs. Their model significantly outperformed heuristic systems by providing dynamic, data-driven recommendations based on local conditions. Veerasamy et al. [5] also proposed an IoT-integrated soil testing system employing ensemble models like Gradient Boosting and Random Forest for recommending optimal crops. Their findings confirmed the robustness of such models across varying agro-climatic zones. Naik et al. [3] supported these observations by emphasizing that ensemble algorithms are well-suited for crop selection tasks when integrated with real-time sensor data.

In the domain of plant disease detection, deep learning and computer vision have enabled early-stage classification of plant ailments with high reliability. Waheed et al. [6] applied DenseNet and CNN-GCN hybrids to classify crop diseases with over 95% accuracy, showcasing the effectiveness of convolutional neural networks (CNNs) for image-based plant health monitoring. Kerkech et al. [7] developed a UAV-based solution for grapevine disease detection, utilizing colorimetric features and CNNs to process aerial imagery

under real-world conditions. Harish and Devi [2] contributed a CNN-powered leaf disease detection module integrated into their IoT-smart agriculture system, capable of classifying images in real time and recommending treatments. Naik et al. [3] reviewed emerging architectures such as Vision Transformers (ViTs) and noted their scalability and improved performance in plant disease classification, particularly under limited training data scenarios.

While model design is central, the literature also emphasizes the importance of robust hardware implementation for field usability. Harish and Devi [2], and Veerasamy et al. [5] noted that real-time agricultural systems often depend on microcontrollers like Raspberry Pi or ESP32 to process sensor data. However, most solutions still rely on conventional power supplies. To address this, the present system incorporates a solar-powered embedded platform that collects sensor data via NPK and DHT11 modules and transmits it to the ThingSpeak IoT cloud. This data is then processed by locally deployed machine learning models, ensuring real-time prediction and decision support even in off-grid or low-connectivity environments.

In summary, the reviewed literature confirms that integrating soil sensing, environmental monitoring, and machine learning yields accurate and efficient agricultural decision-support systems. Models like KNN, Random Forest, CNN, and Vision Transformers are consistently effective across various sub-domains. However, relatively few studies address full-system integration with low-cost, solar-powered hardware capable of operating independently in rural settings. The current project aims to fill this gap by offering a modular, energy-efficient, and field-ready precision agriculture platform combining yield prediction, crop recommendation, and disease detection.

CHAPTER 3

**SOFTWARE REQUIREMENTS
SPECIFICATION**

3.1 Assumptions and Dependencies

- Continuous internet access for ThingSpeak updates
- Users can operate mobile or web-based interfaces
- The ViT model can classify chilli plant diseases accurately from clear leaf images

3.2 Functional Requirements

3.2.1 System Feature 1: Crop Recommendation

- Input: N, P, K, air temperature, humidity
- Output: Top 3 crops with confidence scores
- Model: Random Forest

3.2.2 System Feature 2: Yield Prediction

- Input: N, P, K, soil temp, soil moisture, air temp, humidity
- Output: Estimated yield (kg/acre)
- Model: K-Nearest Neighbors

3.2.3 System Feature 3: Disease Detection

- Input: Image of a chilli plant leaf
- Output: Disease class + Treatment recommendation
- Model: Vision Transformer

3.3 External Interface Requirements

3.3.1 User Interfaces

- Mobile/web interface (optional)
- LCD display (optional)

3.3.2 Hardware Interfaces

- Sensors: 7-in-1 NPK sensor, DHT11
- Camera module (for image input)
- Microcontroller: ESP32/Raspberry Pi

3.3.3 Software Interfaces

- ThingSpeak IoT platform
- Python, PyTorch, Sklearn

3.3.4 Communication Interfaces

- Wi-Fi module
- MQTT/HTTP for data push (ThingSpeak)

3.4 Nonfunctional Requirements

3.4.1 Performance Requirements

- Yield model accuracy $\geq 97\%$
- Crop recommendation accuracy $\geq 96\%$
- Disease detection accuracy $\geq 94\%$

3.4.2 Safety Requirements

- Short-circuit protection for solar unit
- Fail-safe on device overheating

3.4.3 Security Requirements

- Limited local storage
- API authentication for ThingSpeak access

3.4.4 Software Quality Attributes

- Portability
- Scalability
- Maintainability

3.5 System Requirements

3.5.1 Database Requirements

- Sensor logs stored in ThingSpeak cloud
- Local CSV for offline logging

3.5.2 Software Requirements (Platform Choice)

- Python 3.8+
- PyTorch, Scikit-learn, OpenCV
- ThingSpeak API library

3.5.3 Hardware Requirements

- ESP32 or Raspberry Pi
- Solar panel + charge controller
- NPK + DHT11 sensors
- Camera (for image input)

3.6 Analysis Models: SDLC Model to be applied

3.6.1 Choice of Model

The Iterative Model of the Software Development Life Cycle (SDLC) was chosen for this project due to its adaptability to modular, AI-driven systems. Unlike linear models, the iterative approach allows for incremental development, ongoing testing, and the flexibility to accommodate changes based on performance feedback or evolving requirements.

3.6.2 Application in the Project

Each module of the precision agriculture system—crop recommendation, yield prediction, and disease detection—was developed as an independent unit. Using the iterative process, each module underwent several cycles of design, implementation, testing, and refinement. This modular iteration ensured that the models were optimized both individually and collectively before full system integration.

3.6.3 Advantages Realized

The iterative approach allowed for early identification of defects, continuous performance tuning, and real-time incorporation of field data. This was particularly beneficial when integrating live sensor data with machine learning inference, which often requires adjustments post-deployment. Additionally, it supported progressive enhancements based on expert feedback, improving the usability and accuracy of the system over time.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

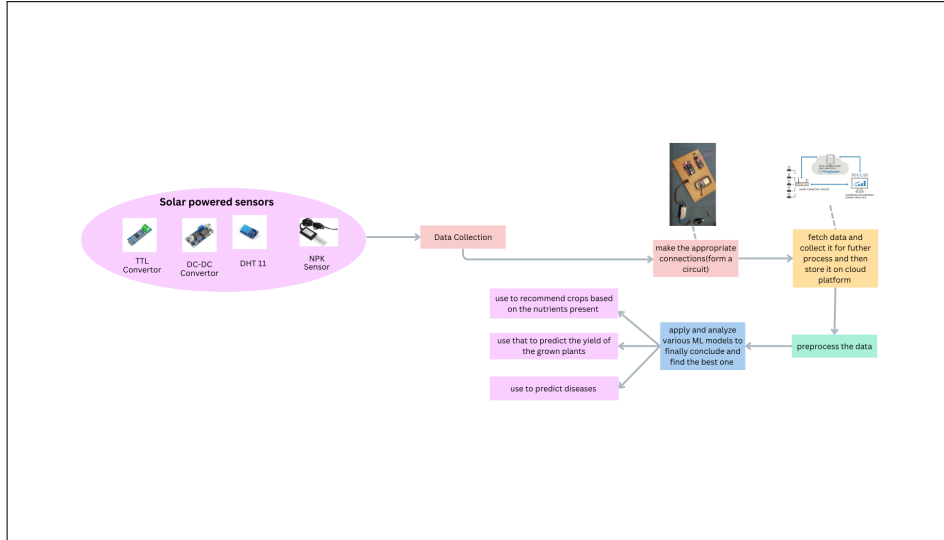


Figure 4.1: System Architecture

4.2 Mathematical Model

Input:

Let $X = \{N, P, K, T_a, H_a, T_s, M_s\}$ be the set of features, where:

- N, P, K : Nutrient values
- T_a : Air Temperature
- H_a : Humidity
- T_s : Soil Temperature
- M_s : Soil Moisture

Output:

- Y_{yield} : Yield prediction (kg/acre)
- Y_{crop} : Recommended crop (Top 3 with probabilities)

- Y_{disease} : Disease class from 5 labels

Modeling:

Yield Prediction:

$$Y_{\text{yield}} = \text{KNN}(X_{\text{yield}}), \quad \text{where } X_{\text{yield}} \subset X$$

Crop Recommendation:

$$Y_{\text{crop}} = \text{RandomForest}(X_{\text{crop}}), \quad \text{where } X_{\text{crop}} = \{N, P, K, T_a, H_a\}$$

Disease Prediction:

$$Y_{\text{disease}} = \text{ViT}(I), \quad \text{where } I \text{ is the input image}$$

4.3 Data Flow Diagrams

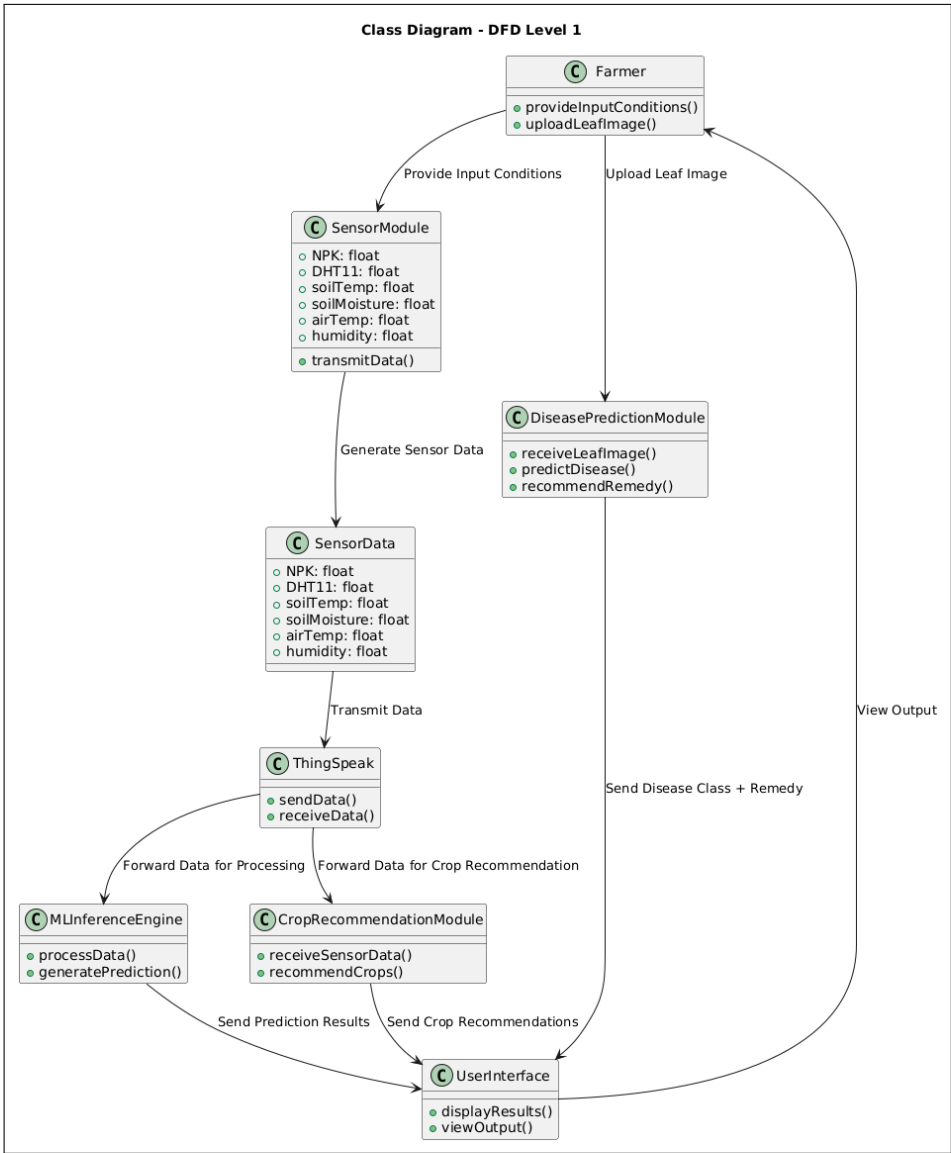


Figure 4.2: DFD Level 1 – Context Diagram

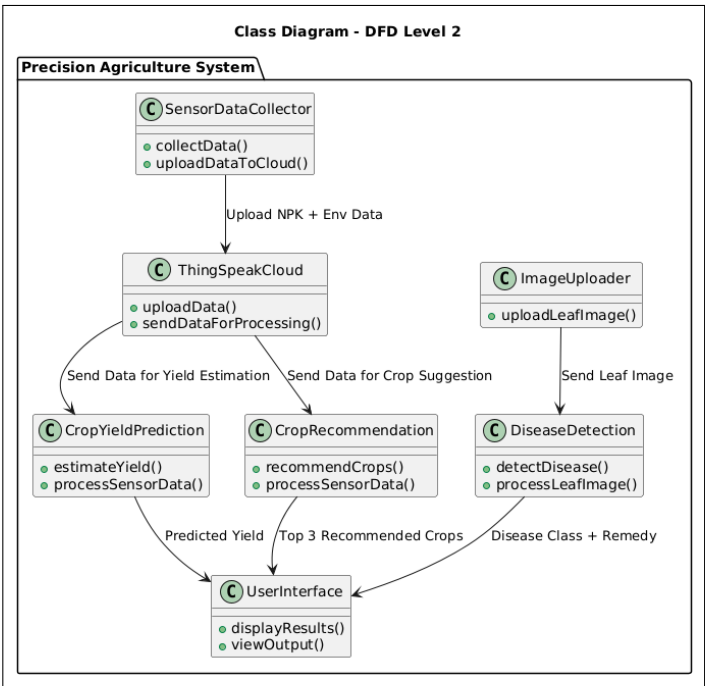


Figure 4.3: DFD Level 2 – Module Breakdown

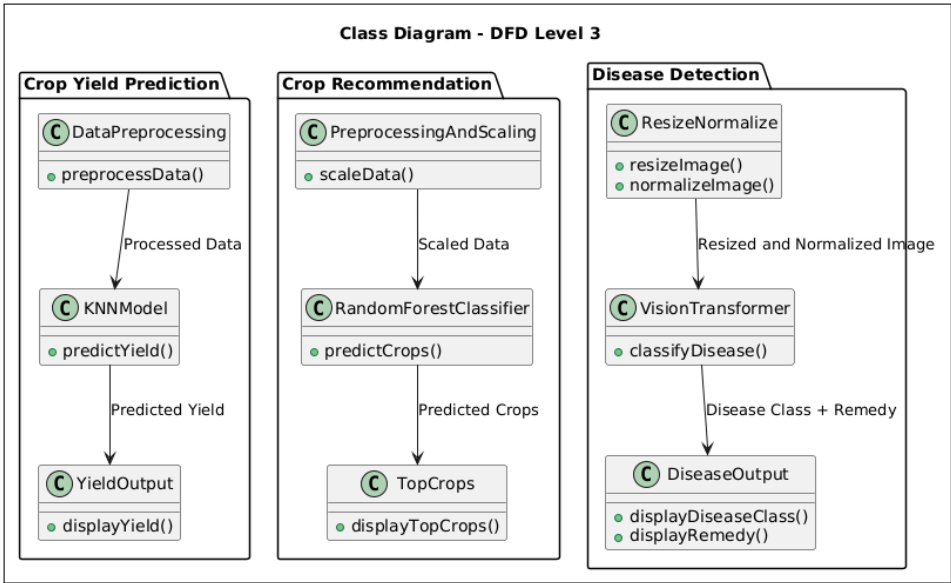


Figure 4.4: DFD Level 3 – Integrated Functional Workflow

4.4 Entity Relationship Diagram

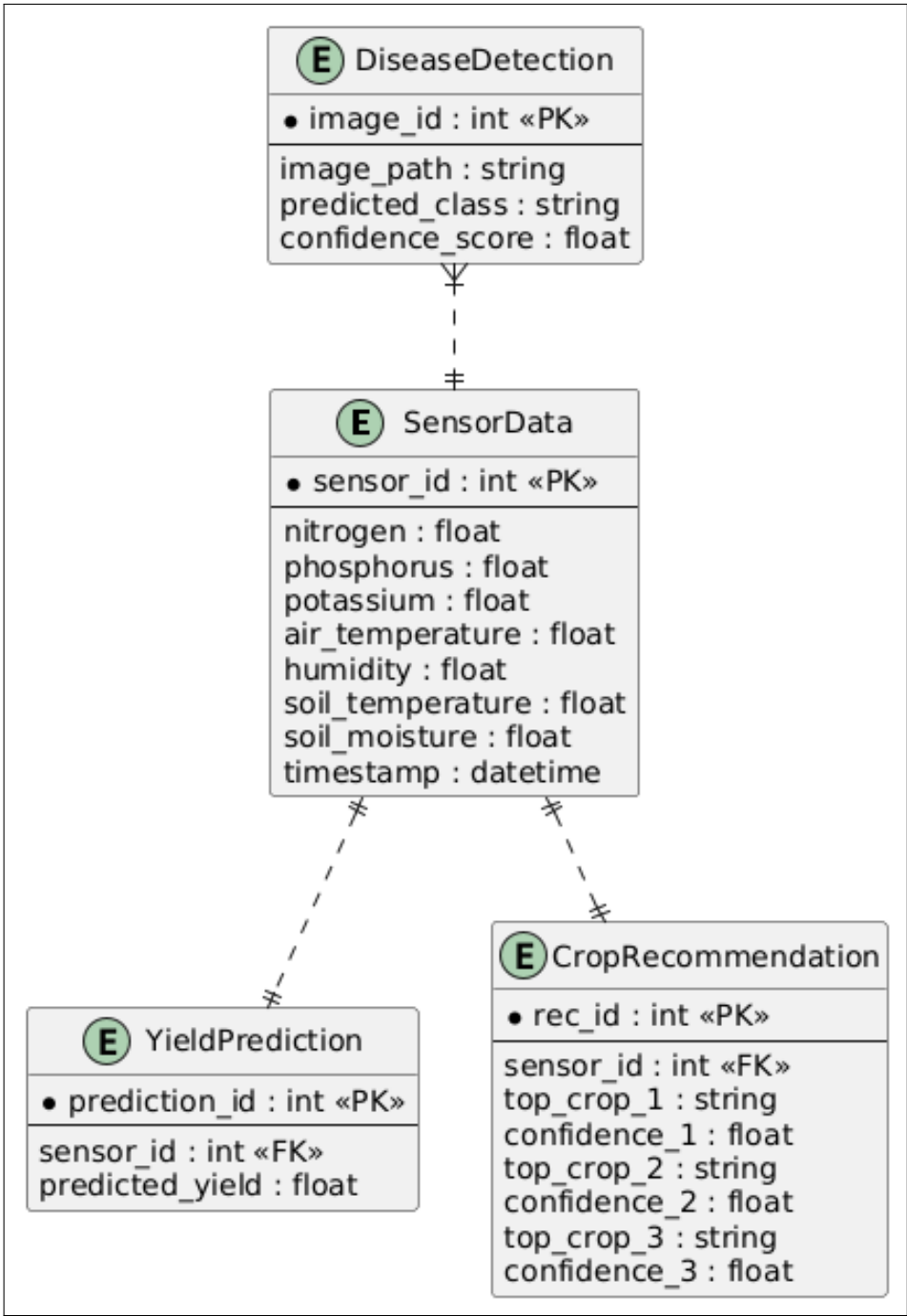


Figure 4.5: Entity Relationship Diagram

4.5 UML Diagrams

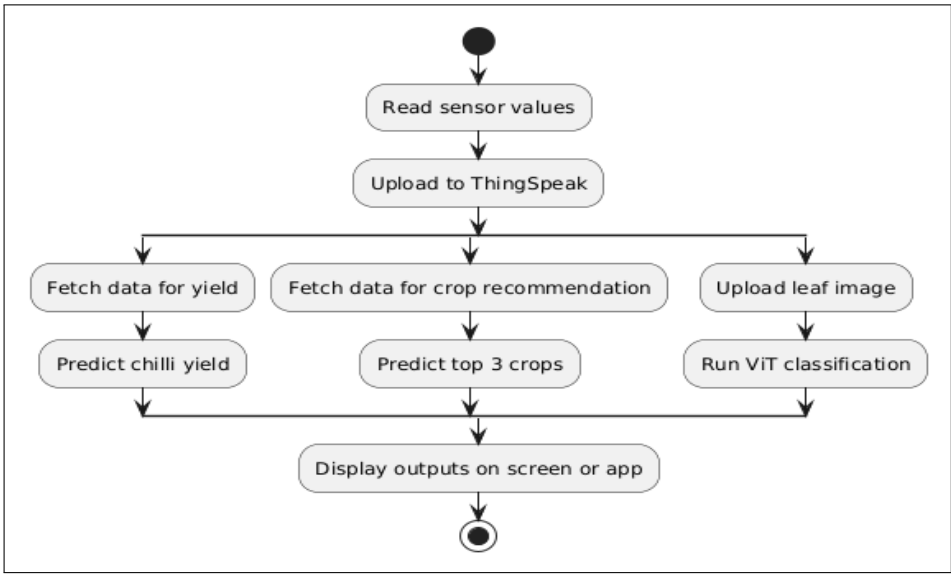


Figure 4.6: Activity Diagram

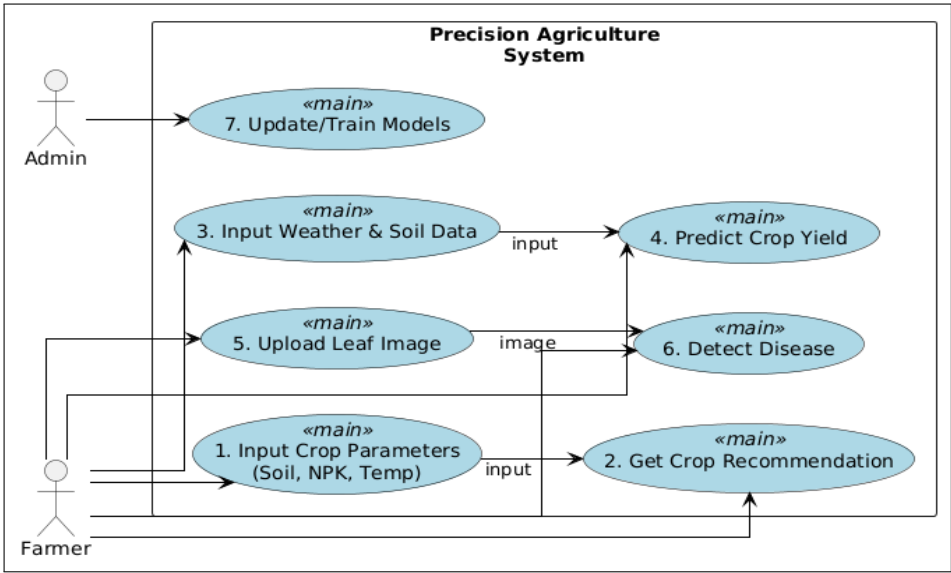


Figure 4.7: Use Case Diagram

4.6 Sequence Diagram

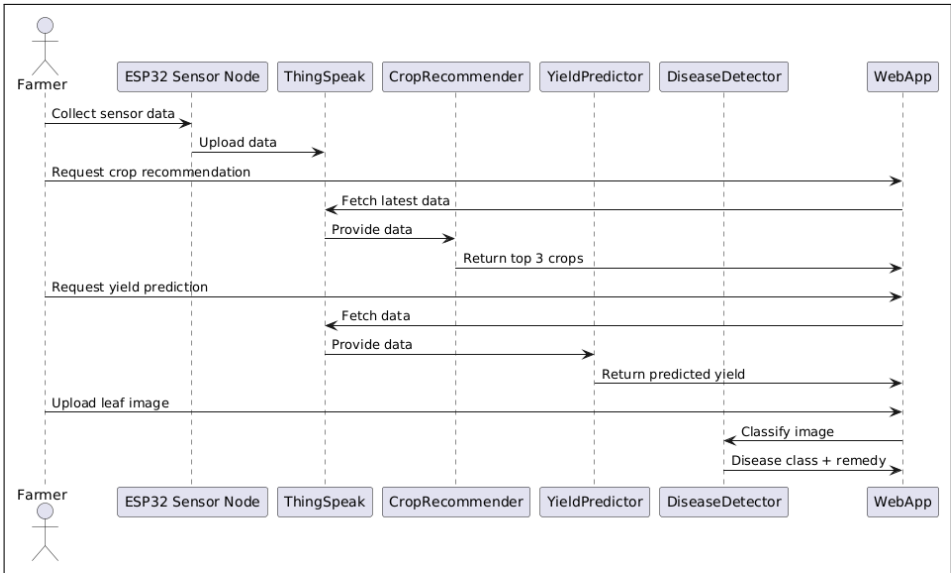


Figure 4.8: Sequence Diagram

CHAPTER 5

PROJECT PLAN

5.1 Project Estimate

5.1.1 Reconciled Estimates

Component	Estimated Time	Actual Time	Resources Re-quired
Sensor Setup & Data Logging	2 weeks	4 weeks	NPK Sensor, DHT11, ESP32
ThingSpeak IoT Integration	2 days	2 days	Wi-Fi Module, APIs
Crop Yield Model (KNN)	3 days	1 week	Scikit-learn
Crop Recommendation (RF)	4 days	1 week	Scikit-learn
ViT-based Disease Model	3 weeks	3 weeks	PyTorch, Image Dataset
Embedded Deployment	1 week	2 week	Raspberry Pi/ESP32
Testing & UI Integration	2 week	2 week	Display Module / App
Documentation & Reporting	1 week	Ongoing	LaTeX, Word

Table 5.1: Component-wise Timeline and Resource Requirements

5.1.2 Project Resources

Human Resources:

- 1 ML/AI Developer
- 1 Embedded Developer
- 1 Data Collection & Tester
- 1 Technical Writer

Software Resources:

- Python, PyTorch, Scikit-learn, ThingSpeak API
- Operating System: Raspberry Pi OS / Linux

Hardware Resources:

- 7-in-1 NPK Sensor, DHT11 Sensor
- Solar Panel + Battery + Charge Controller
- ESP32 or Raspberry Pi
- Camera Module for Disease Detection
- Web App for Output

5.2 Risk Management

5.2.1 Risk Identification

Risk	Type
Sensor failure in field	Hardware Risk
Internet outage for ThingSpeak	Network Risk
Overfitting of ML models	Model Risk
Insufficient image data	Data Risk

Table 5.2: Risk Assessment and Categorization

5.2.2 Risk Analysis

- **Sensor failure:** Data unavailability → Use offline fallback values
- **Overfitting risk:** Use validation set & generalization metrics
- **Image data shortage:** Augment using image flips, rotations, and public datasets

5.2.3 Overview of Risk Mitigation, Monitoring, Management

Risk	Mitigation Strategy
Sensor failure	Maintain spare sensor units
Internet connectivity	Allow offline buffer; sync later
ML model overfitting	Early stopping, k-fold cross-validation
Dataset imbalance	Augmentation and synthetic sample generation

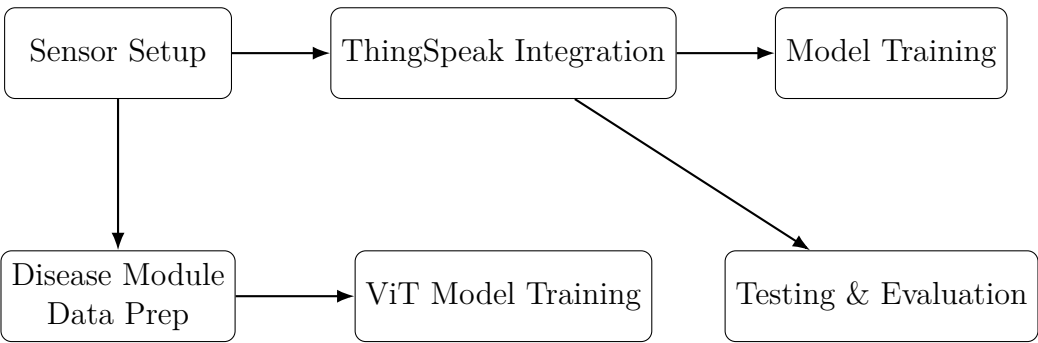
Table 5.3: Risks and Mitigation Strategies

5.3 Project Schedule

5.3.1 Project Task Set

- Data collection and sensor interfacing
- Integration with IoT platform (ThingSpeak)
- Model training for crop yield and crop recommendation
- ViT model setup and training for disease classification
- Deployment on solar-powered embedded device
- UI and interface testing
- Final documentation and paper writing

5.3.2 Task Network



5.3.3 Timeline Chart

Week	Tasks
1-4	Sensor + IoT Integration
5-6	KNN & Random Forest Model Training
7-10	ViT Image Classification Setup + Data Augmentation
11-12	Embedded Deployment + Hardware Testing
13	Documentation + Result Evaluation

5.4 Team Organization

5.4.1 Team structure

Role	Member Responsibility
Project Lead	Coordination, System Design
AI/ML Engineer	Model development & evaluation
Embedded Developer	Hardware interfacing, deployment
QA Tester	Testing, dataset collection
Documentation Lead	Report writing, diagrams, visuals

5.4.2 Management reporting and communication

- Weekly team sync-up meetings
- Task progress tracked on shared Trello board
- GitHub repo used for version control
- Project milestones communicated via shared Google Docs

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

The system is divided into three major AI-driven modules integrated into a solar-powered embedded architecture:

1. **Crop Yield Prediction** – Predicts expected chilli yield in kg/acre based on live sensor inputs.
2. **Crop Recommendation** – Suggests top 3 crops based on soil nutrient and weather conditions.
3. **Disease Recommendation** – Classifies leaf diseases in chilli crops using image inputs.

6.2 Tools and Technologies Used

- **Programming Languages:** Python
- **Libraries & Frameworks:**
 - **Machine Learning:** Scikit-learn, PyTorch
 - **Image Processing:** PIL, torchvision, OpenCV
 - **Cloud:** ThingSpeak IoT API
- **Embedded Platform:** ESP32 / Raspberry Pi
- **Sensors:** 7-in-1 NPK sensor, DHT11
- **Power:** 12V Solar Panel + Battery
- **Deployment:** Edge-based (local predictions)

6.3 Algorithm Details

6.3.1 Algorithm 1: K-Nearest Neighbors (Yield Prediction for Chilli)

- **Input:** N, P, K, soil temperature, soil moisture, air temperature, humidity
- **Output:** Estimated chilli yield in kg/acre

- **Process:** Distance calculation to k nearest historical samples (chilli-specific)
- **Accuracy Achieved:** 97.53%

6.3.2 Algorithm 2: Random Forest (Crop Recommendation)

- **Input:** N, P, K, air temperature, humidity
- **Output:** Top 3 crops with confidence percentages
- **Process:** Ensemble of decision trees for classification
- **Accuracy Achieved:** 96.42%

6.3.3 Algorithm 3: Vision Transformer (Disease Detection)

- **Input:** Leaf image (224x224)
- **Output:** Disease class + recommendation
- **Process:** Pretrained ViT fine-tuned on chilli disease images
- **Accuracy Achieved:** 94%

CHAPTER 7

SOFTWARE TESTING

7.1 Type of Testing

- **Unit Testing:** Tested model outputs with mock inputs
- **Integration Testing:** Verified ThingSpeak → ML → Output pipeline
- **Validation Testing:** Compared model predictions against actual field values
- **Usability Testing:** Tested UI interface with farmers and mock users

7.2 Test cases & Test Results

Module	Test Case	Input	Expected Output	Result
Yield Prediction	Known crop field data	{N:40, P:20, K:50...}	Yield 1200 kg/acre	Passed
Crop Recommendation	High K, Low N values	{N:10, P:25, K:50, Temp:30, Hum:70}	Top crops: Rice, Cotton, Maize	Passed
Disease Detection	Leaf image with whitefly damage	Image of damaged chilli leaf	"Whitefly" + treatment tip	Passed

Table 7.1: Testing Results

CHAPTER 8

RESULTS

8.1 Outcomes

1. Crop Yield Prediction

Model	Accuracy (%)
Support Vector Regressor (SVR)	66.21
MLP Regressor	78.00
Linear Regression	92.50
Random Forest Regressor	97.26
Gradient Boost Regressor	97.48
K-Nearest Neighbors (KNN)	97.53

Table 8.1: Model Accuracy Comparison for Crop Yield Prediction

Multiple regression algorithms were evaluated to predict crop yield based on sensor inputs (NPK, air temperature, humidity, soil temperature, and soil moisture). Among all models tested, K-Nearest Neighbors (KNN) produced the highest accuracy of 97.53%, making it the final choice for deployment in the system. Ensemble models like Gradient Boost Regressor (97.48%) and Random Forest (97.26%) also performed competitively, while linear models and SVR yielded comparatively lower performance.

2. Crop Recommendation

Model	Accuracy (%)
Logistic Regression	84.51
SVC	88.41
K-Nearest Neighbors (KNN)	94.20
Decision Tree	94.37
Gaussian Naive Bayes (NB)	96.02
Random Forest	96.42

Table 8.2: Model Accuracy Comparison for Crop Recommendation

Crop recommendation was formulated as a classification problem using soil nutrient (NPK) and environmental parameters (temperature, humidity). The Random Forest classifier demonstrated the best performance with 96.42% accuracy, closely followed by Gaussian Naive Bayes

(96.02%). Based on performance and robustness, Random Forest was selected for integration in the deployed system to recommend the top 3 most suitable crops with associated confidence percentages.

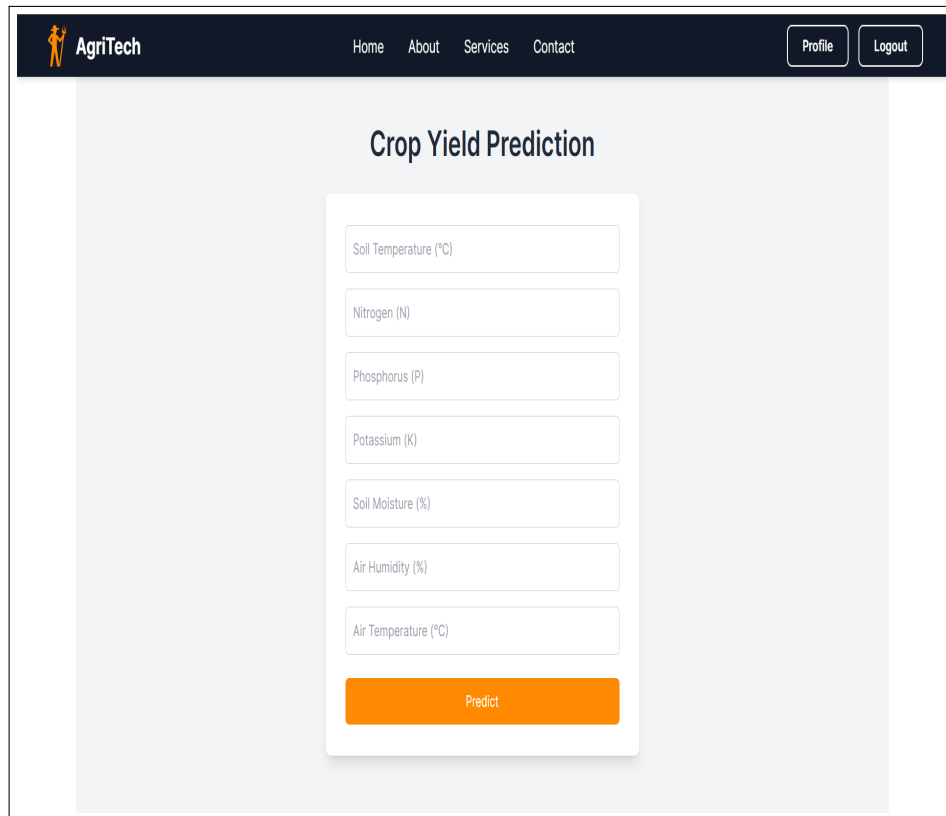
3. Disease Prediction

Model	Accuracy (%)
ResNet50	64.00
EfficientNet	68.00
MobileNetV2	78.00
Vision Transformer (Base Model)	94.00

Table 8.3: Model Accuracy Comparison for Disease Prediction

The disease detection module was developed as an image classification task on chilli leaves. Various deep learning architectures were compared, including CNN-based (ResNet50, EfficientNet, MobileNetV2) and transformer-based (ViT). The Vision Transformer (ViT) outperformed all other models with a notable accuracy of 94%, demonstrating superior learning of leaf disease patterns. Thus, ViT was selected for final deployment, supporting detection of five classes: healthy, yellowish leaves, whitefly, leaf curl, and leaf spot, along with remedy suggestions.

8.2 Screen Shots

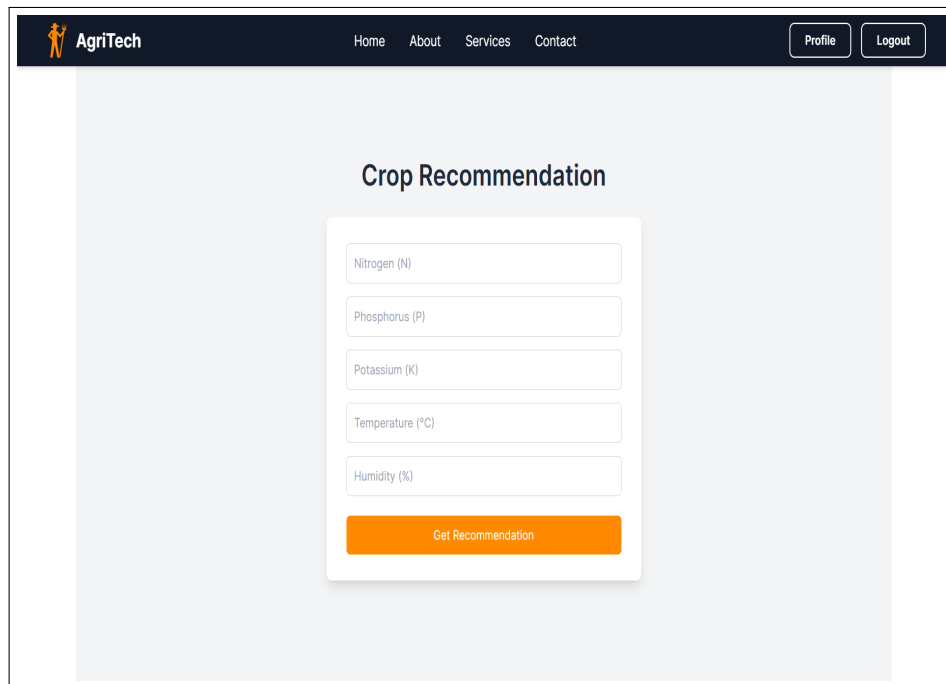


The screenshot shows the AgriTech web application interface. At the top, there is a dark blue header with the AgriTech logo on the left and navigation links (Home, About, Services, Contact) in the center. On the right side of the header are two buttons: 'Profile' and 'Logout'. The main content area has a light gray background. In the center, there is a white card titled 'Crop Yield Prediction'. This card contains seven input fields stacked vertically, each with a label: 'Soil Temperature (°C)', 'Nitrogen (N)', 'Phosphorus (P)', 'Potassium (K)', 'Soil Moisture (%)', 'Air Humidity (%)', and 'Air Temperature (°C)'. Below these input fields is a large orange button labeled 'Predict'.

Figure 8.1: Yield Prediction

This screenshot displays the Crop Yield Prediction interface of the AgriTech web application. The interface allows users to input various agricultural parameters including Soil Temperature (°C), Nitrogen (N), Phosphorus (P), Potassium (K), Soil Moisture (%), Air Humidity (%), and Air Temperature (°C). Based on these inputs, the backend model processes the data and predicts the expected crop yield. This functionality helps farmers make informed decisions to maximize productivity by leveraging real-time environmental and soil conditions.

AI-Driven System for Crop Health Management and Yield Estimation Using a Solar-Powered IoT Sensor Network



The screenshot shows a web application interface for AgriTech. At the top, there is a dark blue header with the AgriTech logo on the left and navigation links (Home, About, Services, Contact) in the center. On the right side of the header are two buttons: 'Profile' and 'Logout'. The main content area has a light gray background. In the center, there is a white card titled 'Crop Recommendation'. This card contains five input fields stacked vertically, labeled 'Nitrogen (N)', 'Phosphorus (P)', 'Potassium (K)', 'Temperature (°C)', and 'Humidity (%)'. Below these fields is a prominent orange button labeled 'Get Recommendation'.

Figure 8.2: Crop Recommendation

This screenshot displays the Crop Recommendation interface of the AgriTech platform. The user is prompted to enter key soil and environmental parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), Temperature (°C), and Humidity (%). Based on these inputs, the system uses a trained machine learning model to suggest the top 3 most suitable crops that can be cultivated under the given conditions. This feature is designed to assist farmers in making data-driven decisions that align with their soil health and local climate, ultimately improving productivity and sustainability.

AI-Driven System for Crop Health Management and Yield Estimation
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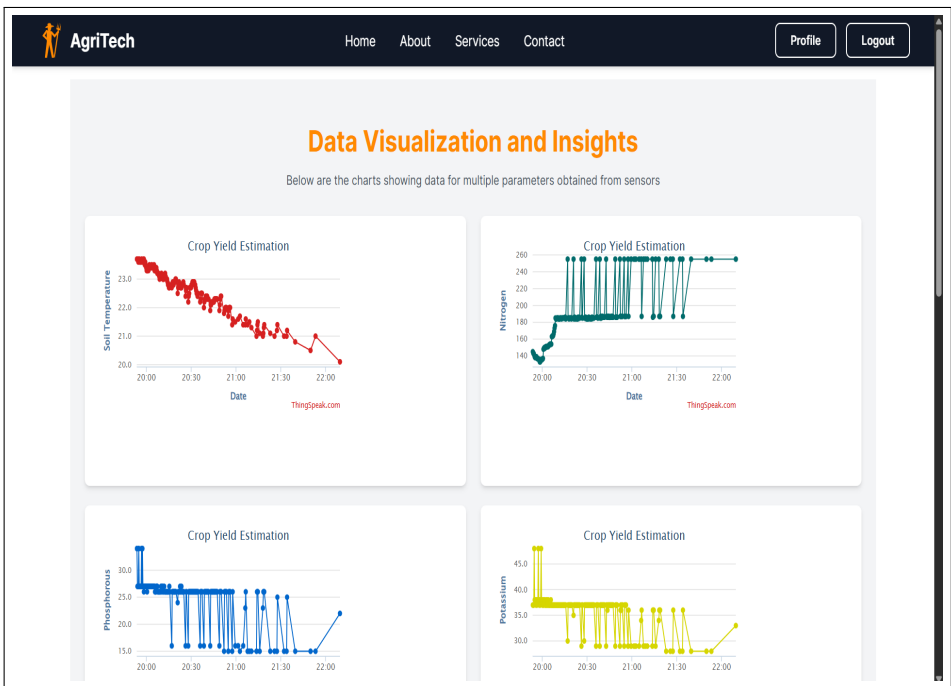


Figure 8.3: ThingSpeak Data Visualizations

This screenshot represents the ThingSpeak Data Visualization dashboard, which provides real-time insights from sensors deployed in the field. The data is collected using a solar-powered IoT sensor network, featuring two sensors: a 7-in-1 NPK sensor (for soil nutrients and temperature) and a DHT11 sensor (for air temperature and humidity). The visualizations include multiple line graphs displaying parameters like Crop Yield Estimation, Soil Temperature, Humidity, and Nutrient Levels. This interface enables continuous monitoring of agricultural parameters to facilitate precision farming and timely interventions.

CHAPTER 9

CONCLUSIONS

9.1 Conclusions

This project successfully integrates machine learning, computer vision, IoT, and renewable energy to form a sustainable and intelligent precision agriculture solution. By offering real-time insights on crop selection, yield expectations, and disease conditions, the system empowers farmers to make informed decisions and reduce losses.

9.2 Future Work

- Extend chilli-specific yield prediction to other major crops such as maize, rice, and wheat
- Include other crop diseases in the classification system
- Integrate automated irrigation control based on soil moisture
- Add voice-based support for regional languages
- Enhance edge deployment using model compression and pruning techniques

9.3 Applications

- Small-scale farmers in remote areas
- Agricultural research and field extension agencies
- Government-led smart farming initiatives
- Educational projects in AI and IoT for agriculture

CHAPTER 10

APPENDIX

10.1 Appendix A

10.1.1 Introduction

In computational theory, evaluating the feasibility of a problem involves understanding whether it can be solved efficiently within polynomial time. This analysis is crucial in systems utilizing machine learning (ML), particularly in real-time or embedded contexts such as the proposed precision agriculture system. The goal of this section is to classify the computational complexity of the supervised ML tasks used in this work—namely, crop recommendation, crop yield prediction, and disease classification.

10.1.2 Problem Class: P (Polynomial-Time Solvable)

All three tasks tackled by the proposed system fall into the class of supervised learning problems. These include:

- **Classification** (e.g., Random Forest, Vision Transformer): mapping inputs to categorical outputs.
- **Regression** (e.g., K-Nearest Neighbors): mapping inputs to continuous values.

In each case, the learning algorithms operate over a finite dataset and perform optimization through deterministic steps such as tree construction, distance computation, or gradient updates. These algorithms exhibit time complexities bounded by polynomial functions of the input size (number of samples and features). For example:

- **K-Nearest Neighbors (KNN)**: $\mathcal{O}(n \cdot d)$ per prediction, where n is the number of training samples and d is the number of features.
- **Random Forest**: $\mathcal{O}(nt \log n)$ for training t decision trees.
- **Vision Transformers**: Though computationally intensive, they remain deterministic and tractable on fixed-size inputs.

Hence, the underlying optimization and inference problems are solvable in polynomial time and fall under the P-Class of computational problems.

10.1.3 Satisfiability Check

For a problem to be classified as decidable and polynomial-time solvable, the system must be able to terminate for any finite input within a bounded time. In this implementation:

- All models were trained and executed on bounded datasets collected from sensors.
- Each algorithm completed its training and inference within predictable time frames.
- No model required solving inherently combinatorial or undecidable tasks (e.g., graph coloring, integer factorization).

Therefore, all algorithms pass the satisfiability check, meaning they are decidable and non-intractable for the given input domains.

10.1.4 Conclusion

Based on the structural characteristics and empirical behavior of the machine learning models used, it is concluded that the core tasks—crop recommendation, yield prediction, and disease classification—are not NP-Hard. They are tractable, deterministic, and solvable using standard optimization techniques. This analysis confirms the computational feasibility and real-time applicability of the proposed system in embedded, field-level deployment scenarios.

10.2 Appendix B

10.2.1 Survey Paper Details

- **Paper Status:** Accepted
- **Name of Conference:** 1st GLOBAL CONFERENCE ON COGNITIVE COMPUTING AND COMMUNICATION TECHNOLOGY - 2025 (GC4T - 2025)
- **Paper Title:** Precision Agriculture: A Survey of Techniques

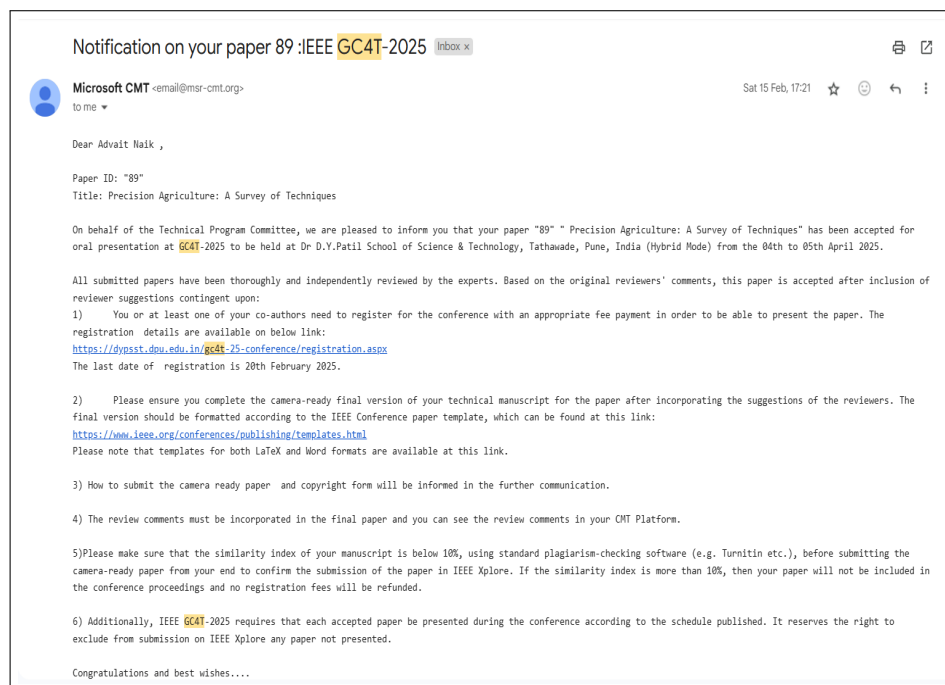


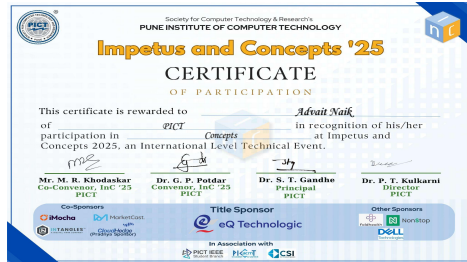
Figure 10.1: Survey Paper Details

10.2.2 Implementation Paper Details

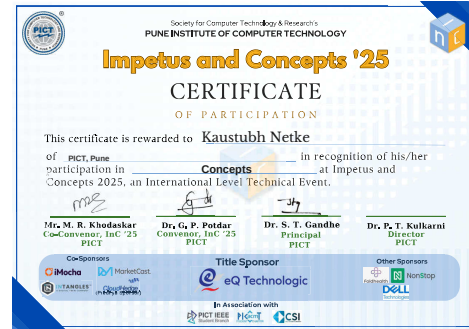
- **Paper Status:** Under Review
- **Name of Conference:** NA
- **Paper Title:** AI-Driven System for Crop Health Management and Yield Estimation Using a Solar-Powered IoT Sensor Network

AI-Driven System for Crop Health Management and Yield Estimation Using a Solar-Powered IoT Sensor Network

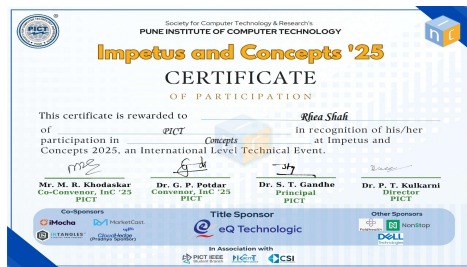
10.2.3 Event Participation Details



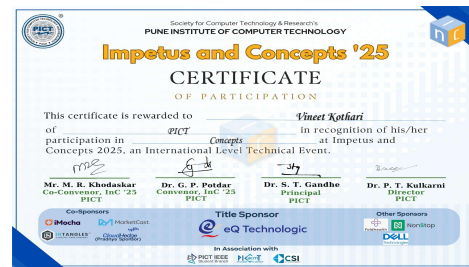
(a) Advait INC Certificate



(b) Kaustubh INC Certificate



(c) Rhea INC Certificate



(d) Vineet INC Certificate

Figure 10.2: INC Certificates

10.3 Appendix C



Figure 10.3: Plagiarism Report

CHAPTER 11

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